FROM MASSES TO AGES

Kaelan Geng

A DISSERTATION

in

School of Natural Sciences

Presented to the Faculties of the University of Galway

in

Partial Fulfillment of the Requirements for the

Degree of Bachelor of Science

2026

Supervisor of Dissertation

Christian Ginski, Full Faculty Title and Affiliation

Graduate Group Chairperson

Graduate Group Chairperson's Typed Name, Full Faculty Title and Affiliation

Dissertation Committee

Committee Member's Typed Name, Full Faculty Title and Affiliation
Committee Member's Typed Name, Full Faculty Title and Affiliation
Committee Member's Typed Name, Full Faculty Title and Affiliation

FROM MASSES TO AGES

COPYRIGHT

2026

Author's Full Legal Name

Write your dedication text in italics here.

ACKNOWLEDGMENT

Write your acknowledgment text here.

ABSTRACT

FROM MASSES TO AGES

Kaelan Geng

Christian Ginski

Write your abstract text here.

TABLE OF CONTENTS

ACKNOWLEDGMENT iv

ABSTRACT v

LIST OF ILLUSTRATIONS vii

CHAPTER 1 : Introduction 1

1.1 Motivation 1

1.2 Formation Mechanisms 2

1.3 Imaging 5

1.4 Evolutionary Models 5

CHAPTER 2 : Methodology 7

2.1 Figures 7

APPENDIX A : TITLE OF APPENDIX A 8

APPENDIX B : TITLE OF APPENDIX B 9

BIBLIOGRAPHY 10

LIST OF ILLUSTRATIONS

FIGURE 1.1 The SU Aurigae system. The majority of this photo comprises of wispy arms, which Ginski et al., 2021 theorise is material from the surrounding medium replenishing the material in the disk. Credit: Ginski et al., 2021 1
FIGURE 1.2 Fig. 3 from Drazkowska2023. This figure provides a general scale for the stages of CA. 3
FIGURE 1.3 Fig 1c of Pollack et al., 1996. The initial rise corresponds to the runaway planetesimal accretion of phase 1; the long plateau corresponds to the transitionary phase 2; the sharp increasing corresponds to both accretion rates rapidly increasing (and the runaway gas accretion) in phase 3 4

CHAPTER 1

Introduction

1.1. Motivation

Determining the age of a planet helps us ascertain the formation delay of the planet, by comparing it to the age of the parent star. We can also test our current formation models, and compare it to the expected disk lifetime (a few Myr, Strom et al., 1993, Manara et al., 2023): if the ages of planets are often younger than the disk lifetimes, then it suggests that disks are being replenished, perhaps through late infall (Kuffmeier et al., 2026). In addition, it helps constrain evolutionary models, assuming we have the planet's mass and luminosity. Normally, we infer a mass estimate from the luminosity and age, while assuming coevality; if we have all 3, we could constrain our evolutionary models accurately enough to determine if the cooling tracks we have are correct.

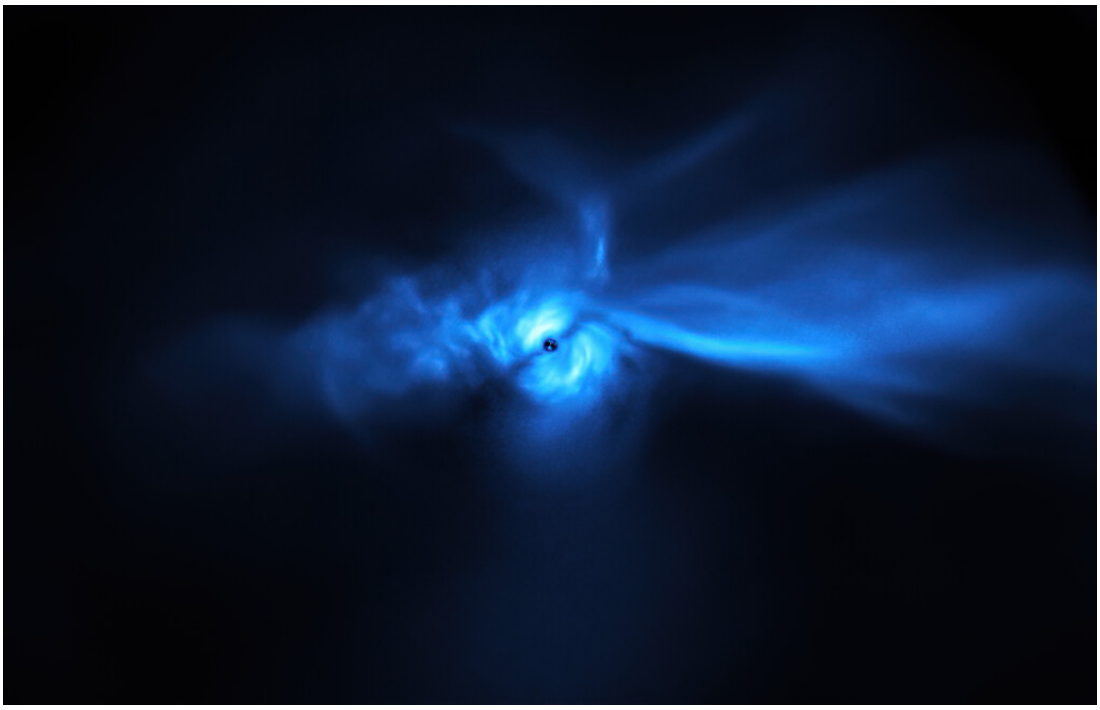


Figure 1.1: The SU Aurigae system. The majority of this photo comprises of wispy arms, which Ginski et al., 2021 theorise is material from the surrounding medium replenishing the material in the disk. Credit: Ginski et al., 2021

One of the major caveats to our current models are that planets and their parent stars are not exactly coeval (Bowler, 2016). Planets can span anywhere from the full stellar age, to 10 Myr younger, depending on the formation mechanism; top-down mechanisms would be close to coeval, while bottom-up mechanisms would leave a discrepancy of a few Myr. Furthermore, depending on the formation mechanism, it could lead to different inferred masses: take for example the young, planetary-mass companion 2M1207-3932 b of the TW Hydrae Association. Assuming coevality, its hot-start mass is $\sim 5M_{Jup}$. But, if the formation is delayed by 8 Myr, then the mass is only $\sim 2.5M_{Jup}$ (Bowler, 2016).

1.2. Formation Mechanisms

1.2.1. Core Accretion

The widely-accepted model for core accretion (CA hereon) comes from Pollack et al., 1996. A general overview of the process is as follows: a protoplanetary disk consists of dust and gas. This dust undergoes random motion, and eventually dust particles may collide and clump together. The repetition of this process eventually leads to pebbles (dust coagulation, Drążkowska et al., 2023), then boulders, plentesimals, then finally, through planetesimal accretion (Safronov and Zvjagina, 1969), a planetary core massive enough to start accreting gas from the disk, which eventually leads to runaway gas accretion (Mizuno, 1980, Bodenheimer and Pollack, 1986, Pollack et al., 1996). All in all, this process' timescale is on the scale of Myr (Pollack et al., 1996), which is significant because a planet formed in this way would be sizeably younger than its host star; planets formed through CA are also what this project aims to find.

This process occurs closer to the centre of the disk than gravitational instability, the other widely-accepted formation pathway, detailed in the next section. Note that this process is not necessarily linear; most pebbles actually don't evolve to the metre-scale. Among the potential barriers include the fragmentation-barrier, drift-barrier and others that are beyond the scope of this project but are detailed in Drazkowska et al. 2023.

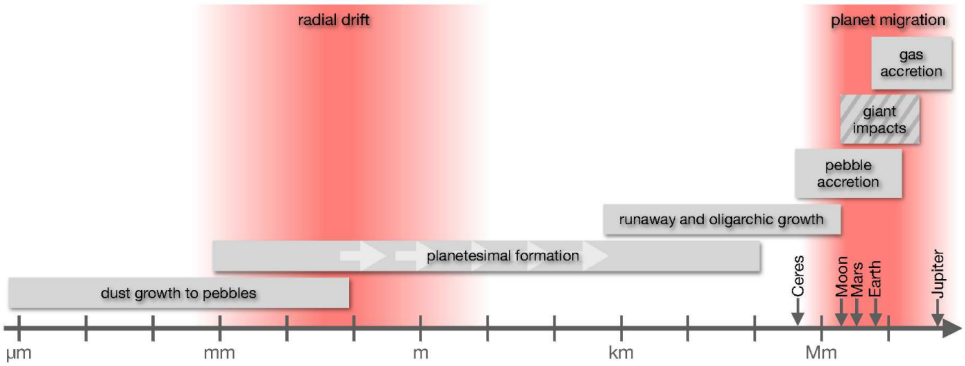


Figure 1.2: Fig. 3 from Drazkowska2023. This figure provides a general scale for the stages of CA.

Previous models that Pollack et al., 1996 model built upon include Mizuno et al., 1978 and Mizuno, 1980 of core instability, Bodenheimer and Pollack, 1986's investigations into Mizuno's models, Pollack et al., 1986 and Podolak et al., 1988 examinations into planetesimal accretion, among others.

In short, a critical mass before runaway gas accretion occurs is found to be $\sim 10M_{\oplus}$ (Mizuno, 1980), later $\sim 10M_{\oplus} - 30M_{\oplus}$ (BP86), and also when the heavy-element mass and the gas mass of the planet would equal; after this critical mass is reached, runaway gas accretion occurs. However, observations of the giant planets also revealed that Neptune and Uranus had compositions such that, in these models, their mass fractions were comprised of too little gas mass, so they'd have never reached the equality required to reach the critical mass. Pollack et al., 1996's remedy to this issue will be found in the outline below.

Arranged neatly, the model of Pollack et al., 1996 is split into three phases: during phase 1, planetesimal accretion exceeds gas accretion rate; they're initially many orders of magnitude apart, with planetesimal accretion rate being the greater value. Overtime, the two rates become comparable: after $\sim 10^5$ years, they are comparable and planetesimal accretion rate begins declining sharply. Meanwhile, gas accretion rate keeps growing. This phase denotes a runaway planetesimal accretion, separate from the runaway gas accretion, and it finishes when a protoplanet has emptied its feeding zone of planetesimals. However, the feeding zone grows as the planet's total mass increases,

introducing new planetesimals (Pollack et al., 1996)

Phase 2 is a novel phase introduced in this model, that denotes a long, liminal phase where both accretion rates are relatively time-invariant, now with gas accretion rate exceeding planetesimal accretion rate. This phase spans the bulk of the aforementioned Myr timescale, with Pollack's simulations of Jupiter lasting $\sim 7Myr$. Finally, phase 3 denotes the runaway gas accretion, where both accretion rates (with gas still far exceeding planetesimal) start rapidly increasing again.

Fig 1c of Pollack et al., 1996 is particluarly relevant to this project: during phase 3, the luminosity grows rapidly to a stage that we can image directly. Once it reaches a maximum luminosity, it will cool down for the rest of its lifetime (since planets have no mechanisms that illuminate them); this cooling is what this project will utilise to infer the age from the luminosity and dynamical masses.

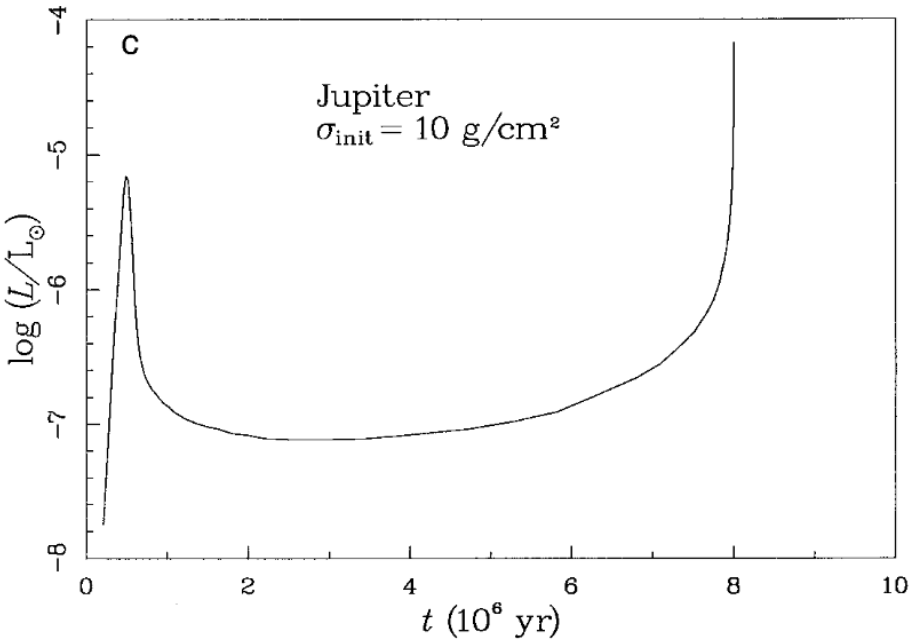


Figure 1.3: Fig 1c of Pollack et al., 1996. The initial rise corresponds to the runaway planetesimal accretion of phase 1; the long plateau corresponds to the transitionary phase 2; the sharp increasing corresponds to both accretion rates rapidly increasing (and the runaway gas accretion) in phase 3

1.2.2. Gravitational Instability

1.2.3. A Note on Cloud Fragmentation and Brown Dwarfs

Cloud fragmentation is traditionally a mechanism that forms stars. A general overview (see Whitworth et al., 2006 for the full process) is that given a molecular cloud is massive enough, it contracts and fragments into subclouds. This process repeats for the subclouds, until eventually the subclouds are so optically thick that they can't contract anymore since they can't radiate away the contraction heat (Whitworth et al., 2006).

The floor of the mass of objects formed seems to reach well into the planetary-mass range ($\sim$ 1 to 4 M_{Jup}, Whitworth and Stamatellos, 2006). Also traditionally, the criterion for brown dwarfs is that it exceeds the deterium burning limit of $\simeq 13 M_{Jup}$ (Chabrier and Baraffe, 2000), however even this remains challenged; Chabrier et al., 2014 argues plantary and brown dwarf can overlap and that there is no clear cut-off mass for one body to be a brown dwarf and another a planet.

Furthermore, the formation tracks for brown dwarfs are also contested: many studies theorise that brown dwarfs can form from the low-end masses of cloud fragmentation (Balmer et al., 2024, Chabrier, 2003). Some others suggest they could also be formed through disk instability (Boss, 1997, Stamatellos et al., 2007, Stamatellos and Whitworth, 2009) or even core accretion along with the associated runaway gas accretion (Balmer et al., 2024, D'Angelo et al., 2010, Mollière and Mordasini, 2012).

1.3. Imaging

1.4. Evolutionary Models

Evolutionary models are split into three categories: hot-start, warm-start and cold-start models. Hot-start start with high entropy retain high entropy after formation, resulting in a large initial radius and luminosity (Bowler, 2016, Bussmann and Helled, 2026). Cold-start models were made prominent by direct imaging to emulate more realistic scenarios for giant planets formed through CA (Bowler, 2016).

Older papers associate cold-start models with core accretion, and hot-start models with disk instability. However, there is a bit more nuance: core accretion can also produce warm or hot initial conditions

CHAPTER 2

Methodology

This project makes use of the species Python package (Stolker et al., 2020). The cooling-curve inversion uses a brute force technique, by using Monte Carlo to sample within the uncertainties of the parameters that are passed in.

2.1. Figures

The captions of figures inside the figure environment (see Figure **??**) will display in the List of Illustrations. Use the `\label` to label figures.[1]

[1]This also includes photographs, maps, schemes, diagrams, etc.

APPENDIX A

TITLE OF APPENDIX A

The content of Appendix A begins here. Use the `\chapter` command to insert additional appendices.

A.1. Section Name of Appendix A

Use the `\section` command to create sections.

APPENDIX B

TITLE OF APPENDIX B

The content of Appendix B begins here.

BIBLIOGRAPHY

William O. Balmer, L. Pueyo, S. Lacour, J. J. Wang, T. Stolker, J. Kammerer, N. Pourré, M. Nowak, E. Rickman, S. Blunt, A. Sivaramakrishnan, D. Sing, K. Wagner, G.-D. Marleau, A.-M. Lagrange, R. Abuter, A. Amorim, R. Asensio-Torres, J.-P. Berger, H. Beust, A. Boccaletti, A. Bohn, M. Bonnefoy, H. Bonnet, M. S. Bordoni, G. Bourdarot, W. Brandner, F. Cantalloube, P. Caselli, B. Charnay, G. Chauvin, A. Chavez, E. Choquet, V. Christiaens, Y. Clénet, V. Coudé Du Foresto, A. Cridland, R. Davies, R. Dembet, A. Drescher, G. Duvert, A. Eckart, F. Eisenhauer, N. M. Förster Schreiber, P. Garcia, R. Garcia Lopez, E. Gendron, R. Genzel, S. Gillessen, J. H. Girard, S. Grant, X. Haubois, G. Heißel, Th. Henning, S. Hinkley, S. Hippler, M. Houllé, Z. Hubert, L. Jocou, M. Keppler, P. Kervella, L. Kreidberg, N. T. Kurtovic, V. Lapeyrère, J.-B. Le Bouquin, P. Léna, D. Lutz, A.-L. Maire, F. Mang, A. Mérand, P. Mollière, C. Mordasini, D. Mouillet, E. Nasedkin, T. Ott, G. P. P. L. Otten, C. Paladini, T. Paumard, K. Perraut, G. Perrin, O. Pfuhl, D. C. Ribeiro, L. Rodet, Z. Rustamkulov, J. Shangguan, T. Shimizu, C. Straubmeier, E. Sturm, L. J. Tacconi, A. Vigan, F. Vincent, K. Ward-Duong, F. Widmann, T. Winterhalder, J. Woillez, S. Yazici, and the GRAVITY Collaboration. VLTI/GRAVITY Provides Evidence the Young, Substellar Companion HD 136164 Ab Formed Like a "Failed Star". *The Astronomical Journal*, 167(2):64, February 2024. ISSN 0004-6256, 1538-3881. doi: 10.3847/1538-3881/ad1689.

Peter Bodenheimer and James B. Pollack. Calculations of the accretion and evolution of giant planets: The effects of solid cores. *Icarus*, 67(3):391–408, September 1986. ISSN 00191035. doi: 10.1016/0019-1035(86)90122-3.

Alan P. Boss. Giant planet formation by gravitational instability. *Science*, 276:1836–1839, 1997.

Brendan P. Bowler. Imaging Extrasolar Giant Planets. *Publications of the Astronomical Society of the Pacific*, 128(968):102001, October 2016. ISSN 0004-6280, 1538-3873. doi: 10.1088/1538-3873/128/968/102001.

Mirco Bussmann and Ravit Helled. Evolutionary tracks of giant planets formed by disk instability. *Astronomy & Astrophysics*, 714:A183, October 2026. ISSN 0004-6361, 1432-0746. doi: 10.1051/0004-6361/202661805.

G. Chabrier, A. Johansen, M. Janson, and R. Rafikov. Giant planet and brown dwarf formation. 2014. doi: 10.2458/azu_uapress_9780816531240-ch027.

Gilles Chabrier. Galactic stellar and substellar initial mass function. *Publications of the Astronomical Society of the Pacific*, 115:763–795, 2003.

Gilles Chabrier and Isabelle Baraffe. Theory of low-mass stars and substellar objects. *Annual Review of Astronomy and Astrophysics*, 38:337–377, 2000.

Gennaro D'Angelo, Richard H. Durisen, and Jack J. Lissauer. Giant planet formation. In S. Seager, editor, *Exoplanets*, pages 319–346. University of Arizona Press, Tucson, 2010.

J. Drążkowska, B. Bitsch, M. Lambrechts, G. D. Mulders, D. Harsono, A. Vazan, B. Liu, C. W. Ormel, K. Kretke, and A. Morbidelli. Planet Formation Theory in the Era of ALMA and Kepler: From Pebbles to Exoplanets, July 2023.

Christian Ginski, Stefano Facchini, Jane Huang, Myriam Benisty, Dennis Vaendel, Lucas Stapper, Carsten Dominik, Jaehan Bae, François Ménard, Gabriela Muro-Arena, Michiel R. Hogerheijde, Melissa McClure, Rob G. Van Holstein, Tilman Birnstiel, Yann Boehler, Alexander Bohn, Mario Flock, Eric E. Mamajek, Carlo F. Manara, Paola Pinilla, Christophe Pinte, and Álvaro Ribas. Disk Evolution Study Through Imaging of Nearby Young Stars (DESTINYS): Late Infall Causing Disk Misalignment and Dynamic Structures in SU Aur*. *The Astrophysical Journal Letters*, 908 (2):L25, February 2021. ISSN 2041-8205, 2041-8213. doi: 10.3847/2041-8213/abdf57.

M. Kuffmeier, A. Winter, A. Kuznetsova, M. Vioque, and A. Gupta. Stellar age is not disk age, September 2026.

C. F. Manara, M. Ansdell, G. P. Rosotti, A. M. Hughes, P. J. Armitage, G. Lodato, and J. P. Williams. Demographics of young stars and their protoplanetary disks: Lessons learned on disk evolution and its connection to planet formation, March 2023.

H. Mizuno. Formation of the Giant Planets. *Progress of Theoretical Physics*, 64(2):544–557, August 1980. ISSN 0033-068X, 1347-4081. doi: 10.1143/PTP.64.544.

H. Mizuno, K. Nakazawa, and C. Hayashi. Instability of a Gaseous Envelope Surrounding a Planetary Core and Formation of Giant Planets. *Progress of Theoretical Physics*, 60(3):699–710, September 1978. ISSN 0033-068X, 1347-4081. doi: 10.1143/PTP.60.699.

P. Mollière and C. Mordasini. Deuterium burning in objects forming via the core accretion scenario: Brown dwarfs or planets? *Astronomy & Astrophysics*, 547:A105, 2012. doi: 10.1051/0004-6361/201219844.

Morris Podolak, James B. Pollack, and Ray T. Reynolds. Interactions of planetesimals with protoplanetary atmospheres. *Icarus*, 73(1):163–179, January 1988. ISSN 00191035. doi: 10.1016/0019-1035(88)90090-5.

James B. Pollack, Morris Podolak, Peter Bodenheimer, and Bruce Christofferson. Planetesimal dissolution in the envelopes of the forming, giant planets. *Icarus*, 67(3):409–443, 1986. ISSN 0019-1035. doi: https://doi.org/10.1016/0019-1035(86)90123-5. URL https://www.sciencedirect.com/science/article/pii/0019103586901235.

James B. Pollack, Olenka Hubickyj, Peter Bodenheimer, Jack J. Lissauer, Morris Podolak, and Yuval Greenzweig. Formation of the Giant Planets by Concurrent Accretion of Solids and Gas. *Icarus*, 124(1):62–85, November 1996. ISSN 00191035. doi: 10.1006/icar.1996.0190.

V.S. Safronov and E.V. Zvjagina. Relative sizes of the largest bodies during the accumulation of planets. *Icarus*, 10(1):109–115, January 1969. ISSN 00191035. doi: 10.1016/0019-1035(69)

90013-X.

D. Stamatellos, D. A. Hubber, and A. P. Whitworth. Brown dwarf formation by gravitational fragmentation of massive, extended protostellar discs. *Monthly Notices of the Royal Astronomical Society*, 382:L30–L34, 2007.

Dimitris Stamatellos and Anthony P. Whitworth. The properties of brown dwarfs and low-mass hydrogen-burning stars formed by disc fragmentation. *Monthly Notices of the Royal Astronomical Society*, 392(1):413–427, 2009. doi: 10.1111/j.1365-2966.2008.14069.x.

Tomas Stolker, Sascha P. Quanz, Kamen O. Todorov, Jonas Kühn, Paul Mollière, Michael R. Meyer, Thayne Currie, Sebastian Daemgen, and Baptiste Lavie. MIRACLES: Atmospheric characterization of directly imaged planets and substellar companions at 4-5 μm. I. Photometric analysis of β Pic b, HIP 65426 b, PZ Tel B and HD 206893 B. *Astronomy & Astrophysics*, 635:A182, March 2020. ISSN 0004-6361, 1432-0746. doi: 10.1051/0004-6361/201937159.

Stephen E. Strom, Suzan Edwards, and Michael F. Skrutskie. Evolutionary Time Scales for Circumstellar Disks Associated with Intermediate-Type and Solar-Type Stars. In *Protostars and Planets III*, page 837, January 1993.

A. P. Whitworth and D. Stamatellos. The minimum mass for star formation, and the origin of binary brown dwarfs. *Astronomy & Astrophysics*, 458(3):817–829, November 2006. ISSN 0004-6361, 1432-0746. doi: 10.1051/0004-6361:20065806.

Anthony Whitworth, Matthew R. Bate, Aake Nordlund, Bo Reipurth, and Hans Zinnecker. PPV Chapter - The Formation of Brown Dwarfs, February 2006.